

Applying Calibrated Explanations: A Primer

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(SOTA)
Machine Learning
(ML) Models**

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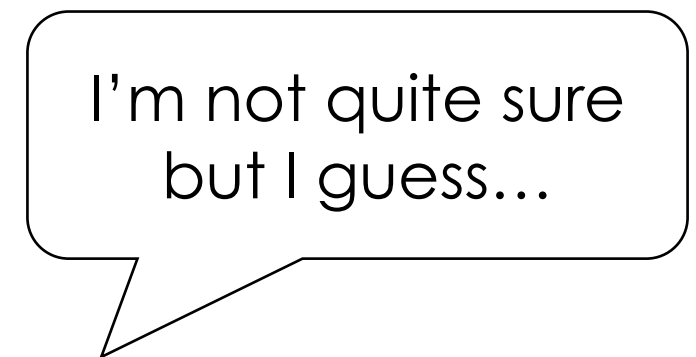
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Furthermore, they do not provide insights on what conditions would change the decision of the model with corresponding confidence score

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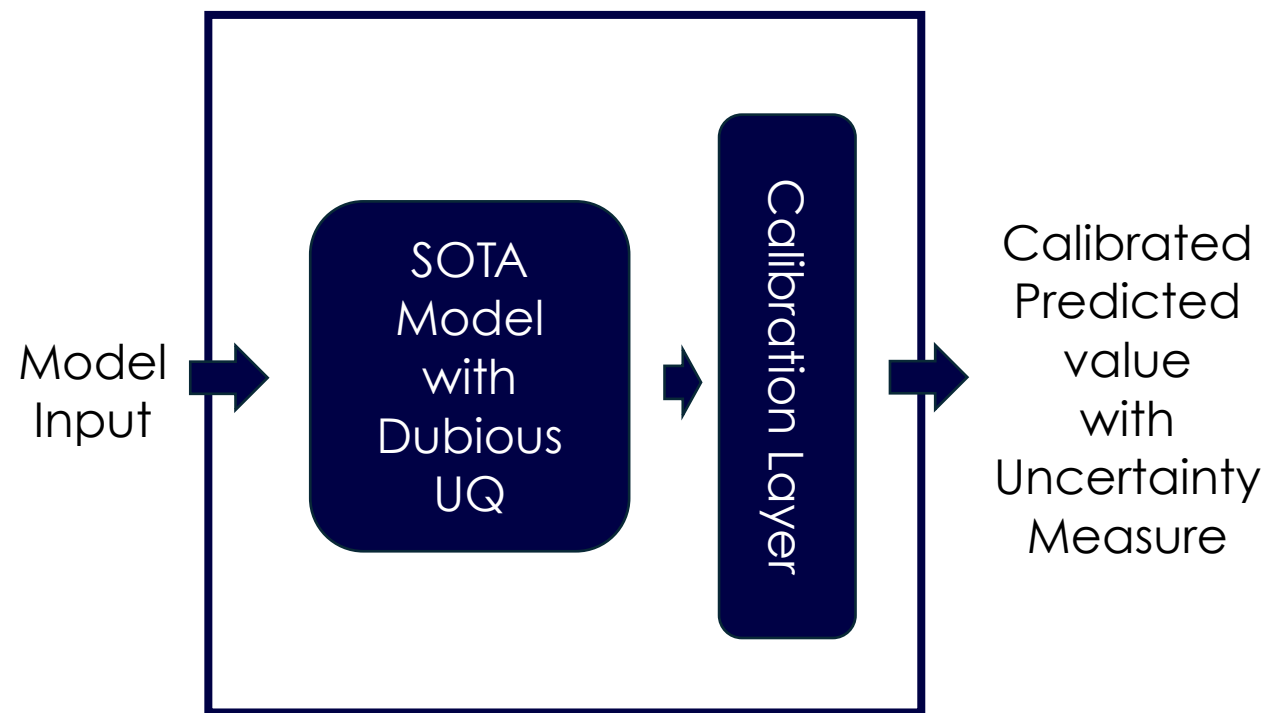


We naturally state how confident we are when answering questions.

Uncertainty Quantification(UQ) gives the model the ability to express "how sure am I?", not just a risk score, but a confidence around it.

- **Wider band of scores:** low-certainty, unconfident
- **Narrower band of scores:** High certainty, communicates confidence in predictions

How It Works: Three-Step Workflow



Many machine learning pipelines miss the calibration part

- Train your usual model with training data.
- **Calibrate** with calibration data to fix over/under-confidence and get intervals.
- The calibrated model now produces a prediction value with uncertainty measure (**risk score with intervals**)

Calibrated Explanations

1. On Predictions :
Decision (with confidence score)
 - How sure is the model?
2. On Explanations :
Drivers of the decision (with confidence score for each driver)
 - Why the model says that, and how confident is each reason?
3. On Counterfactuals :
Factors that could change the decision (with confidence score for each factor)
 - What would change the decision of the model?

Predictions: How to Interpret Outputs from *Calibrated Explanations* Framework

Model Input

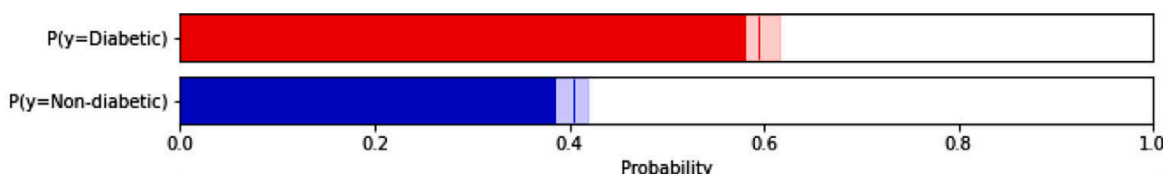


Calibrated ML Model

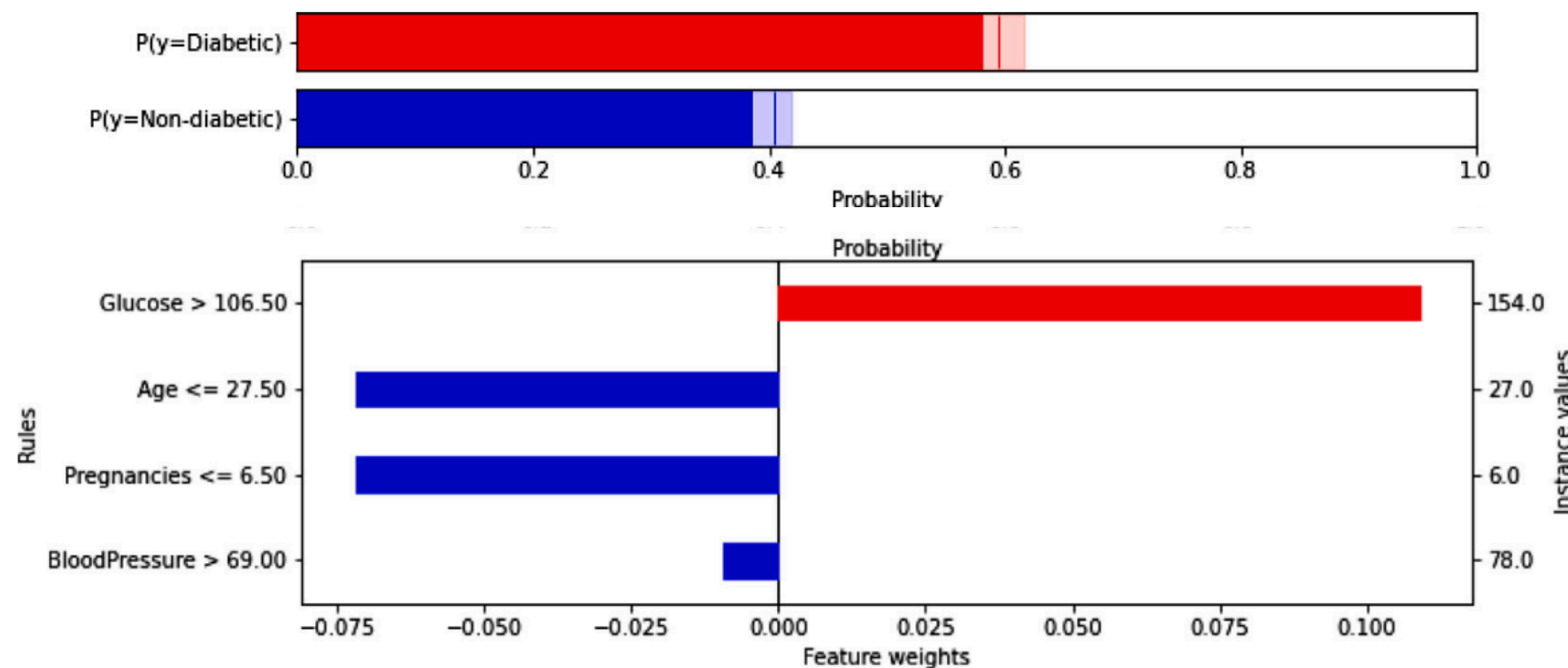


$P(y = \text{Diabetic}) = 0.595$ **[0.582-0.622]***

- A calibrated model produces **prediction intervals**.
- $P(y = \text{Diabetic}) = 59.5\%$ (Point Estimate)
 - Uncertainty band: 58.2% to 62.2%
- $P(y = \text{non-Diabetic}) = 40.5\%$ (Point Estimate)
 - Uncertainty band: 38.7% to 42.0%
- Wide band = be cautious;
- Narrow band = trust more



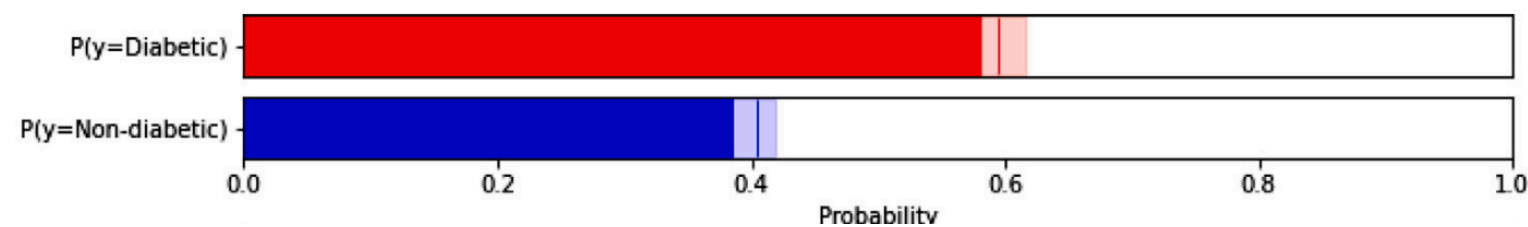
Explanations: How to Interpret Main Drivers of Predictions (First Mode)



- Feature weights describe how each feature **affects the model predictions**
- First mode only highlights **point effects**: which features pushed the risk **up (red) or down (blue)**, and by how much.
- The first mode is similar to existing local explanations (LIME, local SHAP)

This is like reporting a risk value without a reference range: effects are clear but statement of effect can be **overconfident**.

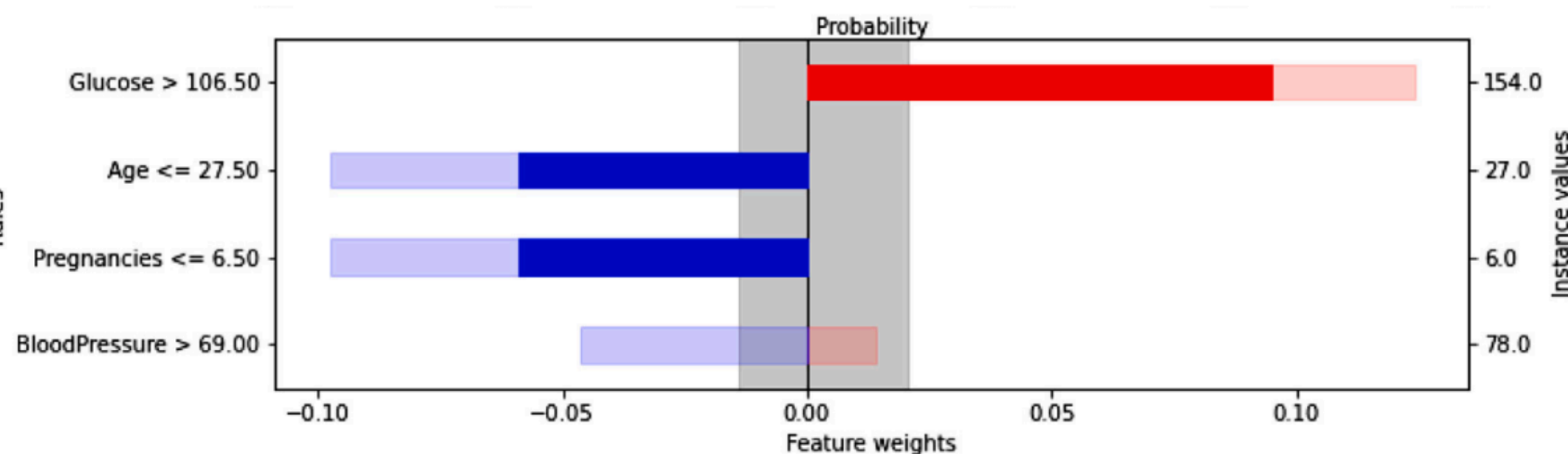
Explanations: How to Interpret Main Drivers of Predictions (Second Mode)



- Second mode provides **feature effect plus their uncertainty bands:**

Dark bar = estimated effect of rule

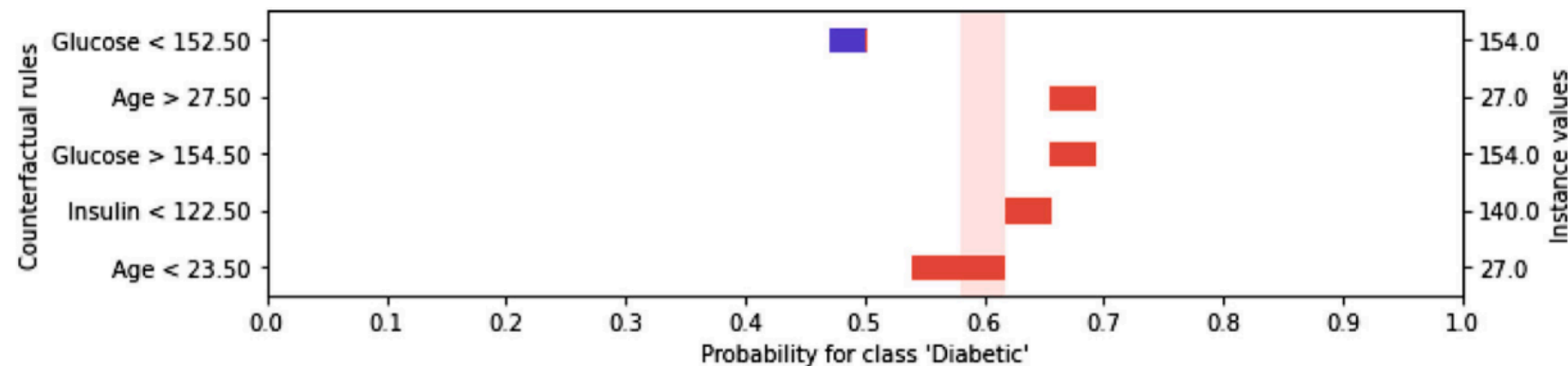
Light halo = plausible range of effect



- **Gray vertical zone around 0.00** = "no clear effect region"; if a band touches

First mode tells us what drove the score; second mode with uncertainty tells us how sure we should be about each driver, so we can act on the solid signals and treat borderline ones indeterminate

Counterfactuals: Actionable “What-Ifs” with Uncertainty



Caveat: counterfactuals are not causal guarantees; they reflect the model’s behavior, not proof that changing a value will change outcomes.

- Right vertical axes: instance values
- Left vertical axes: counterfactual rules
- Plot: Describes the net effect of counterfactual rule, **blue decreases risk, red increases risk**
- **Example:** If Glucose < 152.50, probability would decrease to 0.48-0.50.
- All other rules, **Age >27.50, Glucose > 154.50, Insulin < 122.50, Age < 23.50**, would push the diabetes risk higher if these rules are true.

Calibrated Explanations

1. Calibrated Probability (CP) – How sure is the model?
2. Calibrated Explanations (CE) – Why the model says that, and how confident is each reason?
3. Counterfactual Calibrated Explanations (CCE) – What would change the decision of the model?